



RESPIRATORY PATHOLOGY

PC-I MCQ Exam Study Guide (2017 Exam Edition)

Prepared Under	DR. DOC MEDICAL STUDY SERIES
Course / Module	Respiratory System Pathology (PC-I)
Affiliation	Jimma University Department of Pathology
Content Scope	30 High-Yield Exam Questions with Comprehensive Rationales
Document Version	v2.0 (Official Branded Edition with Custom Watermark)

PREFACE & STUDY INSTRUCTIONS

Welcome to the official **Dr. Doc Pathology Study Series**. This guide has been compiled directly from the official **Jimma University 2017 Respiratory Pathology MCQ Exam**. To maximize your understanding and perform exceptionally in your assessments, this guide does not simply provide answers; it unpacks the pathophysiological mechanics underlying every single question.

Each station is analyzed in high-yield detail, including a structured breakdown of why the correct answer is correct, followed by a detailed choice-by-choice refutation explaining why every other distractor option is incorrect. This methodology fosters active recall and critical diagnostic reasoning, rather than rote memorization.

High-Yield Note: When preparing for respiratory pathology, pay close attention to the anatomical landmarks (e.g., larynx vs. pharynx boundaries), the physiological differences between obstructive and restrictive lung diseases, and the specific genetic/morphological signatures of bacterial and neoplastic conditions.

Question 1: Which of the following statements is false about nasal polyps?

- A. They are focal protrusions of the mucosa due to recurrent attacks of rhinitis
- B. Most arise from the lateral nasal wall or ethmoid recess
- C. Most people with nasal polyps are atopic
- D. When large the polyps may encroach on the airway and impair sinus drainage

Correct Answer: C. Most people with nasal polyps are atopic

Explanation:

• **Why C is correct:** This statement is false. While nasal polyps have histologic features (mucosal edema and loose stroma with eosinophils) suggesting an allergic/atopic origin, the majority of people who develop nasal polyps are actually not atopic. Statistically, only 0.5% of atopic patients eventually develop nasal polyps.

- A is a true statement; polyps are benign focal protrusions of the nasal or sinus mucosa resulting from recurrent, chronic attacks of rhinitis.
- **B is a true statement;** nasal polyps typically arise anatomically from the lateral nasal wall or the ethmoid recess.
- D is a true statement; when polyps grow very large or are multiple, they physically block the nasal airway and impair normal paranasal sinus drainage, predisposing to sinusitis.

Question 2: A six years old female patient presented with sore throat, loss of appetite and low grade fever of 3 days duration. On physical examination vital sign is normal and on oropharyngeal examination, the whole pharynx is reddened along with swollen palatine tonsil but there is no whitish membranous exudates seen. Which of the following is true about the pathology in this patient?

- A. Late sequelae, such as rheumatic fever and glomerulonephritis is expected in such cases
- B. The cause of tonsillo-pharyngitis in this patient is likely bacteria like beta-hemolytic streptococci, and Staphylococcus aureus
- C. Most commonly implicated causes are the rhinoviruses, echoviruses, and adenoviruses
- D. A and B

Correct Answer: C. Most commonly implicated causes are the rhinoviruses, echoviruses, and adenoviruses

Explanation:

• **Why C is correct:** The clinical presentation of mild pharyngeal erythema and swollen tonsils without any whitish follicular membranous exudates, in the setting of low-grade fever, is highly typical of a viral pharyngotonsillitis. The most common etiologic agents of viral sore throats are rhinoviruses, echoviruses, and adenoviruses.

- A and B are incorrect because bacterial pharyngitis (classically caused by Group A Streptococcus pyogenes) is typically associated with high
- grade fever, severe dysphagia, and prominent, thick purulent follicular exudates over the tonsillar surface. Penicillin is given to prevent post
- streptococcal sequelae like rheumatic fever, which is not expected in this mild viral case. Confidential
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Question 3: Which of the following diseases is not a component of Kartagener syndrome?

- A. Chronic Sinusitis
- B. Bronchiectasis
- C. Chronic bronchitis
- D. Dextrocardia

Correct Answer: C. Chronic bronchitis

Explanation:

• **Why C is correct:** Kartagener syndrome (Primary Ciliary Dyskinesia) is a congenital autosomal recessive disorder of ciliary microtubules (dynein arm defect). It is characterized by a classic triad consisting of chronic sinusitis, bronchiectasis, and situs inversus / dextrocardia. Chronic bronchitis is an acquired airway disease clinically defined by persistent productive cough primarily caused by smoking; it is not part of this congenital ciliary syndrome.

- A, B, and D are incorrect because sinusitis (due to impaired ciliary sinus clearance), bronchiectasis (due to chronic necrotizing infections of ciliary
- deficient airways), and dextrocardia (due to loss of ciliary direction during embryonic organ rotation) are the three classic defining components of Kartagener syndrome.

Question 4: Which of the following type of atelectasis is irreversible?

- A. Contraction atelectasis
- B. Compression atelectasis
- C. Resorption atelectasis
- D. All

Correct Answer: A. Contraction atelectasis

Explanation:

• **Why A is correct:** Contraction (cicatrizization) atelectasis occurs when local or generalized pulmonary or pleural fibrosis (scarring) prevents full lung expansion. Because fibrotic tissue is permanent and cannot be expanded or stretched back, contraction atelectasis is classically irreversible.

- B and C are incorrect because compression atelectasis (pleural fluid, air, or blood collapse) and resorption atelectasis (bronchial airway obstruction collapse) are both potentially reversible once the underlying compressive force or obstruction is removed. Confidential
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Question 5: A 9-year-old girl has had a sore throat for the past 2 days. On physical examination there is pharyngeal erythema with yellowish exudates over swollen palatine tonsils. A Gram stain of the exudate shows gram-positive cocci in chains. She is given penicillin therapy. What is the most likely complication prevented by prompt treatment of this girl?

- A. Carditis
- B. Hepatitis
- C. Meningitis
- D. Otitis
- E. Pneumonitis

Correct Answer: A. Carditis

Explanation:

• **Why A is correct:** A tonsillar exudate showing Gram-positive cocci in chains is diagnostic of acute streptococcal tonsillopharyngitis (Group A *Streptococcus pyogenes*). Prompt penicillin therapy is administered specifically to prevent post-infectious autoimmune nonsuppurative sequelae—most notably acute rheumatic fever, which can manifest as life-threatening carditis (rheumatic heart disease).

- B, C, D, and E are incorrect because Group A Streptococcal pharyngitis is not typically associated with post
- infectious autoimmune hepatitis, meningitis, otitis, or pneumonitis as major systemic sequelae prevented by early penicillin treatment.

Question 6: Which of the following statements is true about nasopharyngeal angiofibroma?

- A. It is a highly vascular malignant neoplasm
- B. It is well-delineated tumor with no risk of intracranial extension
- C. Recurrence rates are low with negligible case fatality rate
- D. It occurs almost exclusively in adolescent males

Correct Answer: D. It occurs almost exclusively in adolescent males

Explanation:

• **Why D is correct:** Nasopharyngeal angiofibroma is a rare, highly vascular, and histologically benign stromal tumor that occurs almost exclusively in adolescent males and shows a strong clinical association with Familial Adenomatous Polyposis (FAP).

- A is incorrect because it is histologically benign, not malignant.
- B and C are incorrect because despite being benign, it is locally aggressive and destructive, carrying up to a 20% recurrence rate. It can erode bone to produce intracranial extension and has a 9% case fatality rate due to life threatening hemorrhage.
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Question 7: Which of the following statement is true?

- A. Infection has no role in chronic bronchitis
- B. Asthma is usually fatal
- C. Smoking is common risk factor for mesothelioma
- D. Bronchiectasis more commonly affects the upper lobe of the lung
- E. B and D are correct
- F. None

Correct Answer: F. None

Explanation:

- **Why F is correct:** None of the statements (A, B, C, or D) are true.
- A is incorrect because infection plays a critical secondary role in maintaining inflammation and triggering acute exacerbations in chronic bronchitis.
- B is incorrect because asthma is highly manageable and rarely fatal, except during status asthmaticus.
- **C is incorrect because smoking has no link with malignant mesothelioma;** mesothelioma is caused by asbestos exposure.
- D is incorrect because bronchiectasis predominantly affects the lower lobes of the lungs bilaterally.

Question 8: A 3-year-old child had difficulty breathing for the past 24 hours. On physical examination, the child is febrile and has a harsh cough with prominent inspiratory stridor. The lungs are clear on auscultation. An anterior-posterior neck radiograph shows the steeple sign caused by edema producing loss of normal shoulders on the subglottic larynx. Which of the following organisms is the most likely cause of the child's condition?

- A. Hemophilus influenzae
- B. Human papillomavirus
- C. Parainfluenza virus
- D. Streptococcus, group A

Correct Answer: C. Parainfluenza virus

Explanation:

- **Why C is correct:** A young child with fever, a barking cough, inspiratory stridor, and subglottic narrowing (the classic steeple sign on neck radiograph) has Croup (laryngotracheobronchitis). The most common cause of Croup is the Parainfluenza virus.
- A is incorrect because H. influenzae causes acute epiglottitis, which is an extremely severe airway emergency presenting with high fever, dysphagia, drooling, and a muffled 'hot potato' voice without a barking cough.
- **B is incorrect because HPV types 6 & 11 cause laryngeal papillomatosis.**
- D is incorrect because Group A Strep causes pharyngitis, not croup. Confidential
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Question 9: Which of the following is false about carcinoma of the larynx?

- A. The most common histologic type is adenocarcinoma
- B. It is commonly seen in male chronic smokers
- C. Prior to malignant transformation, epithelial changes frequently regress after smoking cessation
- D. None

Correct Answer: A. The most common histologic type is adenocarcinoma

Explanation:

• **Why A is correct:** This statement is false, making it the correct choice. Approximately 95% of primary laryngeal cancers are typical Squamous Cell Carcinomas (SCC); primary adenocarcinomas of the larynx are extremely rare.

- B and C are incorrect as they are true statements; laryngeal SCC is strongly linked to male chronic smokers, and the pre
- cancerous epithelial hyperplastic/dysplastic changes frequently undergo complete regression upon smoking cessation.

Question 10: From first to last, the morphological changes in lobar pneumonia occur in which correct chronological order?

- A. Grey hepatisation, red hepatisation, congestion, resolution
- B. Red hepatisation, grey hepatisation, congestion, resolution
- C. Congestion, grey hepatisation, red hepatisation, resolution
- D. Congestion, red hepatisation, grey hepatisation, resolution
- E. Congestion, red hepatisation, grey hepatisation, interstitial fibrosis

Correct Answer: D. Congestion, red hepatisation, grey hepatisation, resolution

Explanation:

• **Why D is correct:** Lobar pneumonia classically progresses through four well-defined morphological stages: 1. Congestion (Day 1–2): Vascular congestion, heavy red boggy lung, intra-alveolar fluid with bacteria. 2. Red Hepatization (Day 2–4): Firm liver-like lung, alveoli packed with RBCs, neutrophils, and fibrin. 3. Gray Hepatization (Day 4–8): RBCs disintegrate (lyse), while a gray-brown color develops due to persistent fibrinosuppurative exudate. 4. Resolution (Day 8+): Exudate is enzymatically digested, resorbed, or expectorated, restoring normal lung structure. Confidential - For Student Personal Study and Review Only Page 5 RESPIRATORY PATHOLOGY EXAM GUIDE (2017) JIMMA UNIVERSITY

Question 11: A 64-year-old man has had progressive difficulty hearing, particularly with the left ear, over the past 10 years. Audiometric testing shows that he has a bone conduction type of deafness. CT scan of the head shows no abnormal findings. The patient's brother and mother are similarly affected. What is the most likely diagnosis?

- A. Cholesteatoma
- B. Chondrosarcoma
- C. Otitis media
- D. Otosclerosis

Correct Answer: D. Otosclerosis

Explanation:

• **Why D is correct:** Otosclerosis is a familial metabolic bone remodeling disorder of the otic capsule. It is characterized by abnormal bone deposition in the middle ear about the rim of the oval window, leading to conductive (bone conduction) hearing loss by anchoring and immobilizing the stapes footplate.

- A is incorrect because cholesteatomas are non
- neoplastic cystic squamous lesions associated with chronic otitis media.
- B is incorrect because chondrosarcoma is a malignant cartilage
- producing bone tumor.
- C is incorrect because otitis media is an infectious process with middle ear fluid accumulation.

Question 12: What is the most common type of foregut cyst?

- A. Enteric
- B. Bronchogenic
- C. Esophageal
- D. Gastric

Correct Answer: B. Bronchogenic

Explanation:

• **Why B is correct:** **Bronchogenic cysts** are the most common type of congenital foregut cyst. They arise from abnormal detachments of the primitive foregut, are typically located in the hilum or middle mediastinum, and are lined histologically by respiratory epithelium.

- A, C, and D are incorrect because enteric, esophageal, and gastric cysts are less common types of congenital foregut duplication cysts. Confidential
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Question 13: Which one of the following is not cause of resorption atelectasis?

- A. Foreign body aspiration
- B. bronchial asthma
- C. Tumors
- D. Pneumothorax

Correct Answer: D. Pneumothorax

Explanation:

• **Why D is correct:** **Pneumothorax** (air accumulation in the pleural cavity) is a classic cause of **compression (passive) atelectasis**, not resorption atelectasis.

- A, B, and C are incorrect because resorption (obstruction) atelectasis is directly caused by complete airway blockage preventing air from reaching distal alveoli; foreign bodies, mucous plugging in asthma, and intrabronchial tumors are all classic causes.

Question 14: A 24-year-old primigravida has a fetal ultrasound performed at 18 weeks' gestation that shows a normal amniotic fluid index, but there is a large echogenic region in the right lung. A term infant is delivered with no apparent external anomalies, but soon after birth the infant has respiratory distress. CT imaging shows a normal left lung, but there is an area of opacification in the right lung supplied by a vessel from the aorta. Which of the following is the most likely diagnosis?

- A. Extralobar sequestration
- B. Foregut cyst
- C. Hyaline membrane disease
- D. Oligohydramnios sequence
- E. Tracheoesophageal fistula

Correct Answer: A. Extralobar sequestration

Explanation:

• **Why A is correct:** The presence of non-ventilated, sequestered lung tissue showing an abnormal blood supply directly from the aorta (rather than pulmonary arteries) is diagnostic of Pulmonary Sequestration. Because it presents immediately at birth with neonatal respiratory distress and is separated from the normal pleural covering, it is classified as ****extralobar sequestration**** (which is frequently associated with polyhydramnios or diaphragmatic defects).

- B, C, D, and E do not present with this classic pathognomonic finding of systemic arterial feeding vessels supplying a non
- functional pulmonary mass. Confidential
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Question 15: A 30 year old male with cystic fibrosis present with shortness of breath and production of large amount of foul smelling sputum. What is the most likely cause?

- A. Atelectasis
- B. Asthma
- C. Bronchiectasis
- D. Adenocarcinoma

Correct Answer: C. Bronchiectasis

Explanation:

• **Why C is correct:** Cystic fibrosis is the leading hereditary predisposing cause of severe, generalized bronchiectasis. Abnormally thick, viscid mucous secretions plug the airways and impair clearance, leading to chronic, persistent necrotizing bacterial infections (classically *Pseudomonas aeruginosa*). This destroys the elastic and muscular support of bronchial walls, presenting clinically with ****copious, foul-smelling, purulent sputum****.

- A, B, and D are incorrect because none of these conditions are defined structurally by permanent, chronic necrotic dilation of the bronchial tree presenting with massive amounts of foul
- smelling purulent sputum.

Question 16: Among the following gene which one is mostly affected in small cell lung carcinoma?

- A. RB gene
- B. CDK gene
- C. MYC gene
- D. RAS gene

Correct Answer: A. RB gene

Explanation:

• **Why A is correct:** Small cell lung carcinoma (SCLC) is a highly aggressive, neuroendocrine lung malignancy characterized by near-universal loss-of-function mutations in tumor suppressor genes, with ****RB** (retinoblastoma) gene inactivation** and TP53 mutations occurring in close to 100% of cases.

- B and C are incorrect as CDK mutations or MYC amplification (L
- MYC) occur in SCLC but are not as universally present as RB loss.
- D is incorrect because
- RAS
- mutations (such as
- KRAS
-) are characteristic of lung adenocarcinomas but are exceptionally rare in SCLC. Confidential
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Question 17: Which of the following clinical features are not seen in patients with emphysema?

- A. barrel chest
- B. prominent hyperventilation
- C. cyanosis
- D. hunched-over positioning

Correct Answer: C. cyanosis

Explanation:

• **Why C is correct:** Emphysema patients are classically characterized as *****pink puffers*****. Because of severe airspace enlargement and prominent dyspnea, they over-ventilate to maintain relatively normal blood gas values and adequate oxygenation until very late in the disease. Therefore, ****cyanosis is not seen**** in early/moderate emphysema. In contrast, chronic bronchitis patients are *****blue bloaters***** who develop early hypoxia and cyanosis.

- A, B, and D are incorrect because emphysema patients characteristically present with a barrel chest (due to lung over expansion), prominent hyperventilation, and sit forward in a hunched over position to assist with expiration.

Question 18: A 55-year-old man with hematemesis has endoscopic treatment for bleeding esophageal varices. He also has advanced lung disease which would be defined pathologically by 'abnormal permanent enlargement of the airspaces distal to the terminal bronchiole accompanied by destruction of their walls without obvious fibrosis'. What is the single most likely diagnosis?

- A. Alpha 1-antitrypsin deficiency
- B. Cystic fibrosis
- C. Primary biliary cirrhosis
- D. Sarcoidosis
- E. Wilson disease

Correct Answer: A. Alpha 1-antitrypsin deficiency

Explanation:

• **Why A is correct:** The pathological definition describes severe **panacinar emphysema**. When panacinar emphysema is co-existent with hepatic cirrhosis (presenting as portal hypertension and bleeding esophageal varices), the underlying cause is **alpha-1 antitrypsin (A1AT) deficiency**. This genetic deficiency leads to unregulated protease activity destroying alveolar walls and A1AT glycoprotein accumulation causing hepatocyte damage.

- B, C, D, and E are incorrect because none of these conditions are defined pathologically by panacinar emphysema coupled with cirrhotic portal hypertension. Confidential
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Question 19: Which of the following is not typical feature of asthma?

- A. Reversibility
- B. Asthma is recurrent
- C. The severity of asthma usually increase with age
- D. Asthma is usually disabling than lethal

Correct Answer: C. The severity of asthma usually increase with age

Explanation:

• **Why C is correct:** This is not a typical feature, making it the correct exception. Atopic asthma typically begins in childhood and shows an episodic course. Rather than increasing in severity with age, up to **50% of childhood asthma remits spontaneously** during adolescence.

- A, B, and D are incorrect because they are classic hallmarks of asthma; airway obstruction is reversible, attacks are recurrent, and the disease is typically a disabling chronic nuisance but is rarely lethal.

Question 20: Which of the following immune cells is most important in excessive inflammatory response to allergens in persons with atopic asthma?

- A. B lymphocyte
- B. Natural killer cell
- C. TH1 lymphocyte
- D. TH2 lymphocyte

Correct Answer: D. TH2 lymphocyte

Explanation:

• **Why D is correct:** Atopic asthma is driven by a CD4+ helper T-cell response dominated by **TH2 lymphocytes**. These cells secrete cytokines (IL-4, IL-5, IL-13) that stimulate B cells to produce IgE, activate locally recruited eosinophils, and promote mucus hypersecretion.

- A, B, and C are incorrect because B cells, NK cells, and TH1 lymphocytes do not coordinate the allergen
- sensitization or late
- phase eosinophilic reactions characteristic of atopic asthma. Confidential
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Question 21: A 49-year-old man has had increasing dyspnea for the past 4 years. He has an occasional cough with minimal sputum production. On physical examination, his lungs are hyperresonant with expiratory wheezes. Pulmonary function tests show increased total lung capacity (TLC) with slightly increased FVC and decreased FEV1 and FEV1/FVC ratio. Which of the following disease processes should most often be suspected as a cause of these findings?

- A. Centrilobular emphysema
- B. Chronic pulmonary embolism
- C. Diffuse alveolar damage
- D. Nonatopic asthma
- E. Sarcoidosis

Correct Answer: A. Centrilobular emphysema

Explanation:

- **Why A is correct:** A decreased FEV1 and FEV1/FVC ratio (<0.7) accompanied by increased TLC (air trapping) is diagnostic of **obstructive lung disease**. Dyspnea, hyperresonant lungs on percussion, expiratory wheezes, and minimal sputum point directly to severe **centrilobular emphysema** (classically associated with a history of cigarette smoking).
- B, C, and E are incorrect as pulmonary embolism, DAD, and sarcoidosis are restrictive/vascular processes that show decreased TLC on PFTs.
- D is incorrect because nonatopic asthma typically presents with acute episodic attacks rather than a steadily progressive, 4
- year insidious course of dyspnea.

Question 22: Diffuse alveolar damage is the pathologic manifestation of which clinical entity?

- A. Acute respiratory distress syndrome (ARDS)
- B. Pneumonia
- C. Pulmonary hemorrhage
- D. Pulmonary edema
- E. Pulmonary emboli

Correct Answer: A. Acute respiratory distress syndrome (ARDS)

Explanation:

• **Why A is correct:** ****Diffuse Alveolar Damage (DAD)**** is the definitive histopathologic hallmark of ****Acute Respiratory Distress Syndrome (ARDS)****. In the acute stage, it is characterized by capillary congestion, interstitial edema, and the lining of alveolar walls by waxy hyaline membranes.

- B, C, D, and E are incorrect because none of these conditions are defined primarily or specifically by diffuse, bilateral alveolar
- capillary damage forming diagnostic hyaline membranes. Confidential
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Question 23: Asbestos exposure results in all of the following changes except?

- A. Interstitial pneumonia
- B. Malignant mesothelioma
- C. Pleural effusion
- D. Pleural fibrosis

Correct Answer: A. Interstitial pneumonia

Explanation:

- **Why A is correct:** Asbestos exposure leads to chronic, diffuse **interstitial fibrosis (asbestosis)**, which belongs to the restrictive pneumoconioses. **Interstitial pneumonia** is an infectious/inflammatory pathology caused by atypical pathogens (e.g., *Mycoplasma*), not a physical dust-induced fibrotic reaction.
- B, C, and D are incorrect because malignant mesothelioma (strongly associated with asbestos), parietal pleural plaques (fibrosis), and recurrent pleural effusions are all well
- established, direct clinical consequences of asbestos inhalation.

Question 24: A 72-year-old woman has had difficulty with vision in her right eye for 3 months. She also has pain in the right upper chest. The findings on physical examination include unilateral enophthalmos, miosis, anhidrosis, and ptosis on the right side of her face. A chest radiograph shows right upper lobe opacification and bony destruction of the right first rib. Which of the following conditions is most likely to be present in her?

- A. Bronchopneumonia
- B. Bronchiectasis
- C. Bronchogenic carcinoma
- D. Tuberculosis

Correct Answer: C. Bronchogenic carcinoma

Explanation:

• **Why C is correct:** The patient's unilateral enophthalmos, ptosis, miosis, and anhidrosis represent **Horner's syndrome**. An apical lung lesion (**Pancoast tumor**) invading the cervical sympathetic plexus and chest wall causing first rib destruction is typically a **bronchogenic carcinoma** (usually non-small cell lung cancer, SCC or adenocarcinoma).

- A, B, and D are incorrect because bronchopneumonia, bronchiectasis, and tuberculosis do not typically present as a localized, aggressive apical mass that destroys adjacent ribs and causes Horner's syndrome.
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Question 25: The Azzopardi effect is seen in which lung neoplasm?

- A. Squamous cell carcinoma
- B. Adenocarcinoma
- C. Small cell carcinoma
- D. Bronchioloalveolar carcinoma

Correct Answer: C. Small cell carcinoma

Explanation:

• **Why C is correct:** The **Azzopardi effect (phenomenon)** refers to the basophilic lacing and staining of venule and capillary walls by DNA released from necrotic, fragile tumor cells. This classic pathologic finding is highly characteristic of **small cell lung carcinoma (SCLC)** due to its extreme cell turnover and fragility.

- A, B, and D are incorrect because squamous cell carcinoma, adenocarcinoma, and bronchoalveolar carcinoma are non
- small cell lung cancers that do not typically manifest the Azzopardi phenomenon.

Question 26: False about Obstructive airway diseases?

- A. Total lung capacity and forced vital capacity (FVC) are normal
- B. FEV1: FVC ratio is reduced
- C. They are characterized by reduced expansion of lung parenchyma
- D. They are caused by partial or complete obstruction at any level

Correct Answer: C. They are characterized by reduced expansion of lung parenchyma

Explanation:

• **Why C is correct:** This statement is false, making it the correct choice. Reduced expansion of lung parenchyma accompanied by decreased total lung capacity is the definition of **restrictive** lung disease, not obstructive airway disease.

- A, B, and D are incorrect because they are true statements; obstructive airway diseases are characterized by normal TLC/FVC, a reduced FEV1:FVC ratio (<0.7), and are caused by increased resistance to airflow due to partial or complete obstruction at any level. Confidential
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Question 27: Which of the following is the most common site for the development of malignant mesothelioma?

- A. Pleura
- B. Peritoneum
- C. Pericardium
- D. Tunica vaginalis of testis
- E. Mediastinum

Correct Answer: A. Pleura

Explanation:

• **Why A is correct:** **Malignant mesothelioma** is a rare, highly aggressive tumor arising from mesothelial cells. Although it can arise in the peritoneum, pericardium, or tunica vaginalis, it most commonly (approximately 80% or more of cases) develops in the **pleura** (parietal or visceral pleura).

- B, C, D, and E are incorrect because they represent much less common sites of origin for this asbestos
- related neoplasm.

Question 28: A patient receives cardiopulmonary resuscitation suffers aspiration of gastric contents. Over the next 10 days he develops a non-productive cough along with a fever (37.9°C). A chest radiograph reveals a 4 cm diameter mass with an air-fluid level in the right lung. A culture done for the sputum shows mixed flora. Which of the following conditions is he most likely to have?

- A. squamous cell carcinoma
- B. Lung abscess
- C. Chronic bronchitis
- D. Bronchiectasis

Correct Answer: B. Lung abscess

Explanation:

• **Why B is correct:** Aspiration of gastric contents containing mouth flora and gastric acid is the leading cause of a **lung abscess**. The clinical triad of fever, cough, sputum showing mixed flora, and a chest X-ray revealing a **cavity with an air-fluid level** in the right lung (due to the more vertical right main bronchus) is highly diagnostic.

- A, C, and D are incorrect because none of these conditions present as an acute (10 day), localized suppurative cavity containing an air
- fluid level following an aspiration event. Confidential
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Question 29: Which of the following lung cancers is mostly associated with hypercalcemia?

- A. Squamous cell carcinoma
- B. Adenocarcinoma
- C. Small cell carcinoma
- D. Carcinoid tumor

Correct Answer: A. Squamous cell carcinoma

Explanation:

• **Why A is correct:** **Squamous cell lung carcinoma (SCC)** is the histologic subtype most frequently associated with paraneoplastic **hypercalcemia**, caused by the ectopic secretion of parathyroid hormone-related peptide (PTHrP).

- B is incorrect because adenocarcinomas are associated with HPOA/digital clubbing and migratory thrombophlebitis.
- C is incorrect because small cell carcinomas secrete ACTH (Cushing's) or ADH (SIADH).
- D is incorrect because carcinoid tumors rarely secrete serotonin, causing carcinoid syndrome.

Question 30: Restrictive lung disease are characterized by all of the following except?

- A. Shortness of breath
- B. Cyanosis
- C. Tachypnea
- D. Wheezing

Correct Answer: D. Wheezing

Explanation:

• **Why D is correct:** This is the correct exception. **Wheezing** is a high-pitched expiratory whistling sound caused by bronchial narrowing and bronchospasm, which is the hallmark of **obstructive** airway diseases (such as asthma). Restrictive lung diseases feature stiff lungs with decreased compliance and present instead with dyspnea, dry cough, tachypnea, and late cyanosis.

- A, B, and C are incorrect because shortness of breath, tachypnea, and cyanosis are all typical clinical findings seen in restrictive pulmonology. Confidential
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